8257 Direct Memory Access Controller (DMAC) MPCA Module 4

> **Definition:** The **Intel 8257** is a 4-channel programmable **Direct Memory Access Controller (DMAC)** that enables high-speed data transfer directly between I/O peripherals and system RAM without CPU intervention, bypassing the slow CPU fetch-execute cycle.

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1. Detailed Technical Explanation

1. The DMA Concept & Speed Advantage

In standard programmed I/O, every byte transferred from a disk/network card to RAM requires multiple CPU instructions (`IN AL, DX` followed by `MOV [SI], AL`). DMA bypasses the CPU:

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PROG
```

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2. 8257 Internal Block Diagram & Channels

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8257 D
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3. DMA Transfer Sequence (Step-by-Step)

1. Peripheral asserts **DMA Request (`DRQ`)** to 8257.

2. 825

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4. DMA Transfer Modes

1. **Single Transfer Mode:** 8257 transfers one byte per request, releasing the bus back to CPU between transfers.

2. **Bl

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5. Must-Write Points for Exams

- 8257 has **4 independent DMA channels** (Channel 0 to Channel 3).

- Eac

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6. Quick Recall Flow

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DRQ -
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